GOOGLE COLAB OUTPUTS

```
downloaded: ptbxl_database.csv
copied from /content: scp_statements.csv
Metadata folder: /content/ptbxl_subset

All records: 21799 | patients: 18869
Labelled records: 21388
abnormal
abnormal    12319
normal       9069
Name: count, dtype: int64


No patient appears in more than one split.
       records  abnormal_rate
split
test      1500          0.581
train     7000          0.570
val       1500          0.562
```

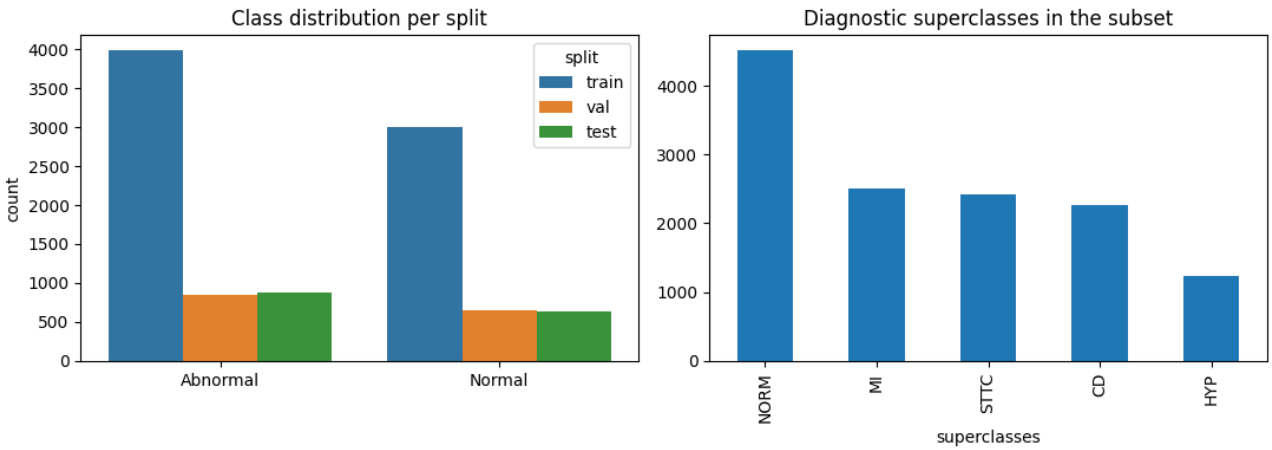
100%

Downloaded 9986 / 10000 records in 1843s; failed: 14

100%

```
 10000/10000 [01:27<00:00, 142.96it/s]
Unreadable records skipped: 14
Saved: /content/drive/MyDrive/PRG4206_Arrhythmia_Project/
data/features_II_7000_1500_1500.csv (9986, 24)
24 features: ['II_mean', 'II_std', 'II_min', 'II_max',
'II_range', 'II_rms'] ...
train/val/test: 6989 1498 1499


Exploratory data analysis
```

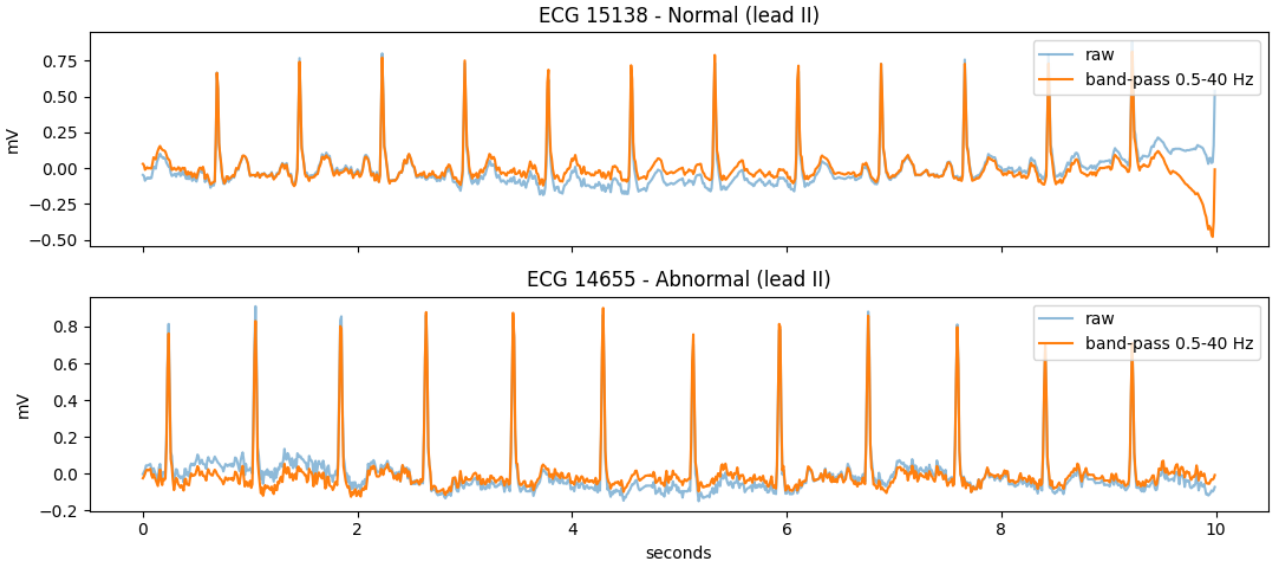



```
Missing values in train (fraction):
II_hr_mean     0.04
II_rr_mean     0.04
II_rr_std      0.04
II_rmssd       0.04
II_pnn50       0.04
dtype: float64
```

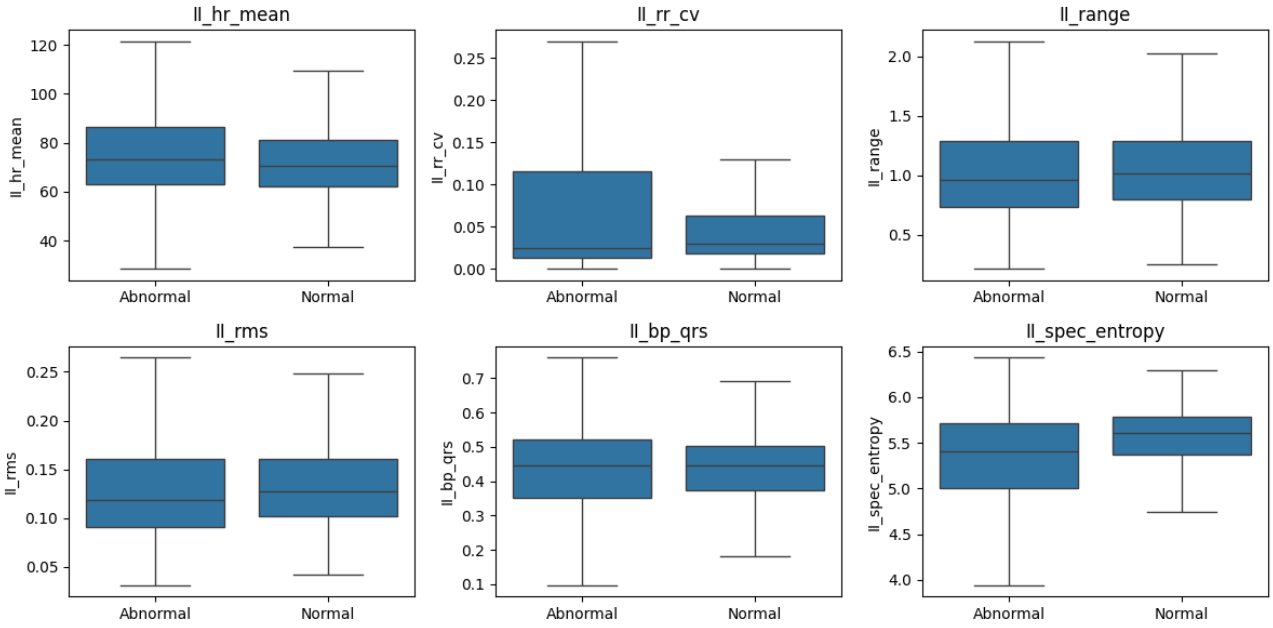
	count	mean	std	min	25%	50%	75%	max
II_mean	6989.0	0.000	0.004	-0.056	-0.001	0.000	0.002	0.027
II_std	6989.0	0.134	0.057	0.031	0.095	0.123	0.160	0.736
II_min	6989.0	-0.349	0.265	-2.693	-0.423	-0.264	-0.179	-0.063
II_max	6989.0	0.737	0.381	0.055	0.462	0.681	0.944	3.154
II_range	6989.0	1.086	0.488	0.215	0.760	0.988	1.290	5.847
II_rms	6989.0	0.134	0.057	0.031	0.095	0.123	0.160	0.738
II_energy	6989.0	0.021	0.022	0.001	0.009	0.015	0.026	0.544
II_skew	6989.0	2.123	1.988	-4.689	1.024	2.577	3.595	6.523
II_kurtosis	6989.0	12.346	7.750	-0.881	6.289	11.480	17.155	64.351
II_bp_baseline	6989.0	0.366	0.148	0.028	0.259	0.345	0.452	0.944
II_bp_qrs	6989.0	0.433	0.112	0.053	0.362	0.446	0.513	0.820
II_bp_high	6989.0	0.190	0.097	0.002	0.122	0.185	0.252	0.664
II_dom_freq	6989.0	3.377	1.954	0.391	1.953	3.125	4.297	16.406
II_spec_centroid	6989.0	8.995	2.272	2.272	7.575	9.064	10.495	18.227
II_spec_entropy	6989.0	5.423	0.475	3.179	5.187	5.514	5.754	6.434
II_n_beats	6708.0	12.463	3.262	3.000	10.000	12.000	14.000	30.000
II_hr_mean	6708.0	75.169	19.587	13.201	62.762	71.817	84.052	182.773
II_rr_mean	6708.0	0.857	0.278	0.328	0.714	0.835	0.956	4.545
II_rr_std	6708.0	0.073	0.155	0.000	0.013	0.024	0.066	3.585
II_rmssd	6708.0	0.105	0.271	0.000	0.014	0.025	0.081	7.170

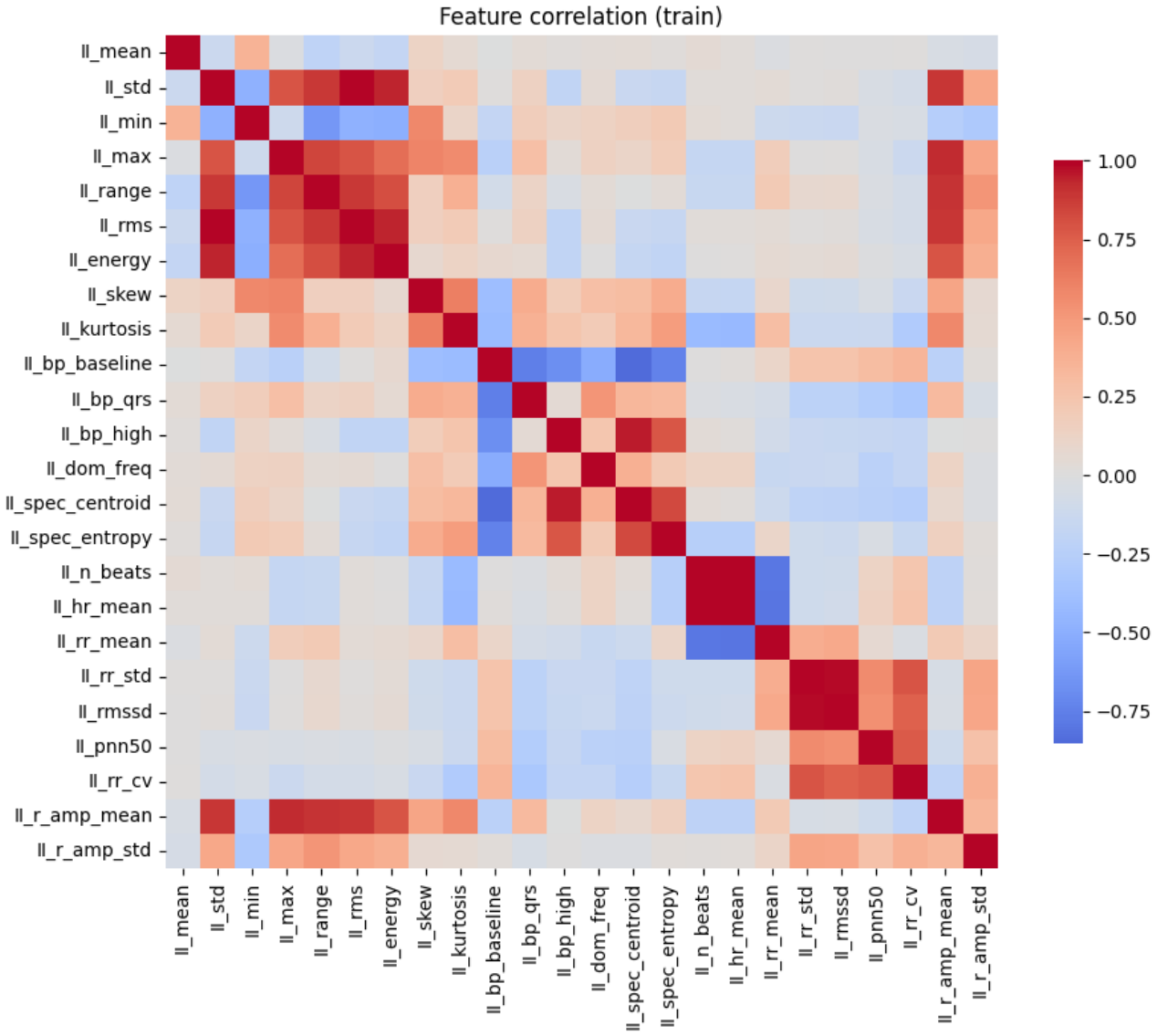
II_pnn50	6708.0	0.230	0.319	0.000	0.000	0.000	0.385	1.000
II_rr_cv	6708.0	0.086	0.131	0.000	0.015	0.028	0.080	1.408
II_r_amp_mean	6708.0	0.682	0.331	0.085	0.440	0.628	0.862	2.801
II_r_amp_std	6708.0	0.049	0.030	0.006	0.032	0.043	0.058	0.485

Before / after preprocessing (raw lead II vs 0.5-40 Hz band-pass



BOXPLOT






```
Logistic Regression  validation AUC = 0.7905 | {'clf__C': 10}
Random Forest        validation AUC = 0.8303 |
{'clf__max_depth': None, 'clf__min_samples_leaf': 5,
'clf__n_estimators': 200}
XGBoost              validation AUC = 0.8378 |
{'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 400}
```

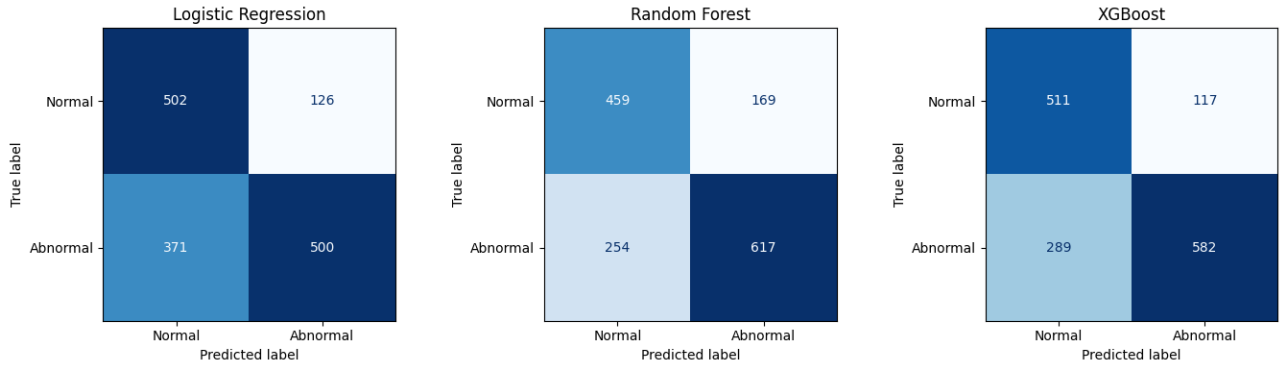
	CV ROC-AUC mean	CV ROC-AUC std
Model		
Logistic Regression	0.7759	0.0141
Random Forest	0.8251	0.0148

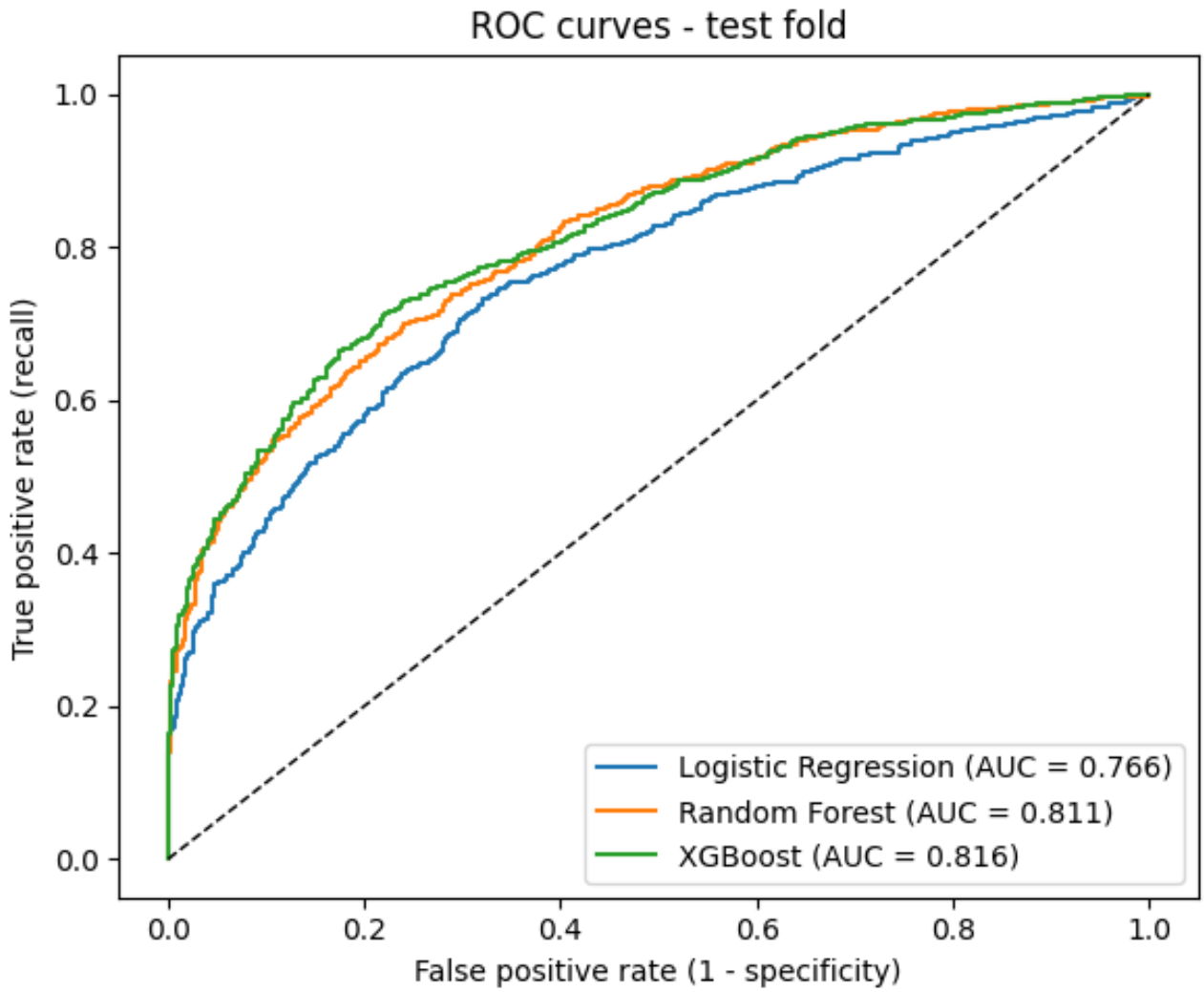
XGBoost	0.8301	0.0102

Evaluation on the untouched test fold (fold 10)

	Accuracy	Precision	Recall (sensitivity)	Specificity	F1	ROC-AUC	AUC 95% CI	PR-AUC
Model								
Logistic Regression	0.6684	0.7987	0.5741	0.7994	0.668	0.7661	0.741-0.790	0.8339
Random Forest	0.7178	0.7850	0.7084	0.7309	0.744	0.8106	0.789-0.83	0.8669
XGBoost	0.7292	0.8326	0.6682	0.8137	0.741	0.8161	0.795-0.83	0.8731

Confusion matrices





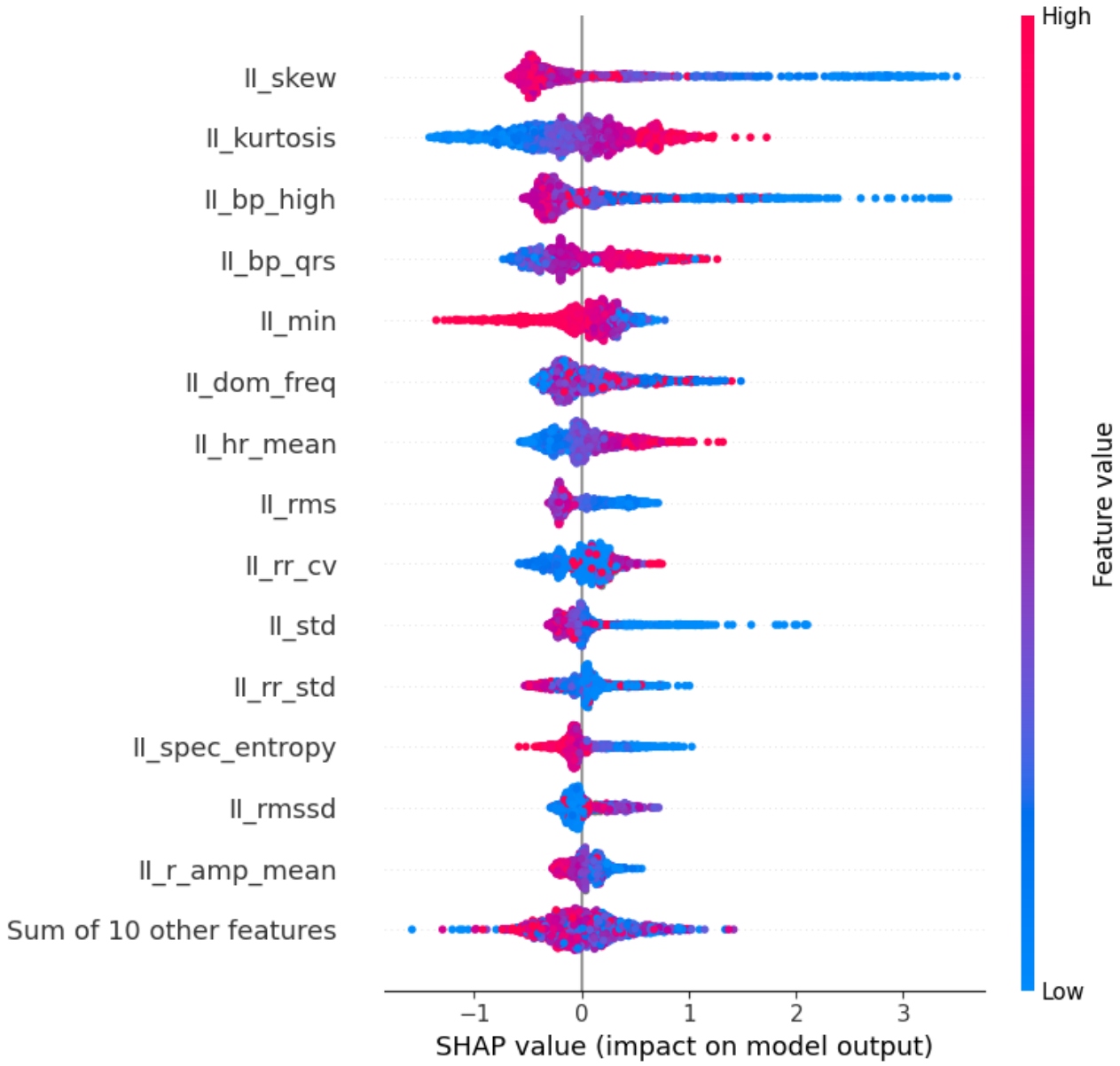
Best model by validation AUC: XGBoost
Recall per diagnostic superclass (records can carry several):

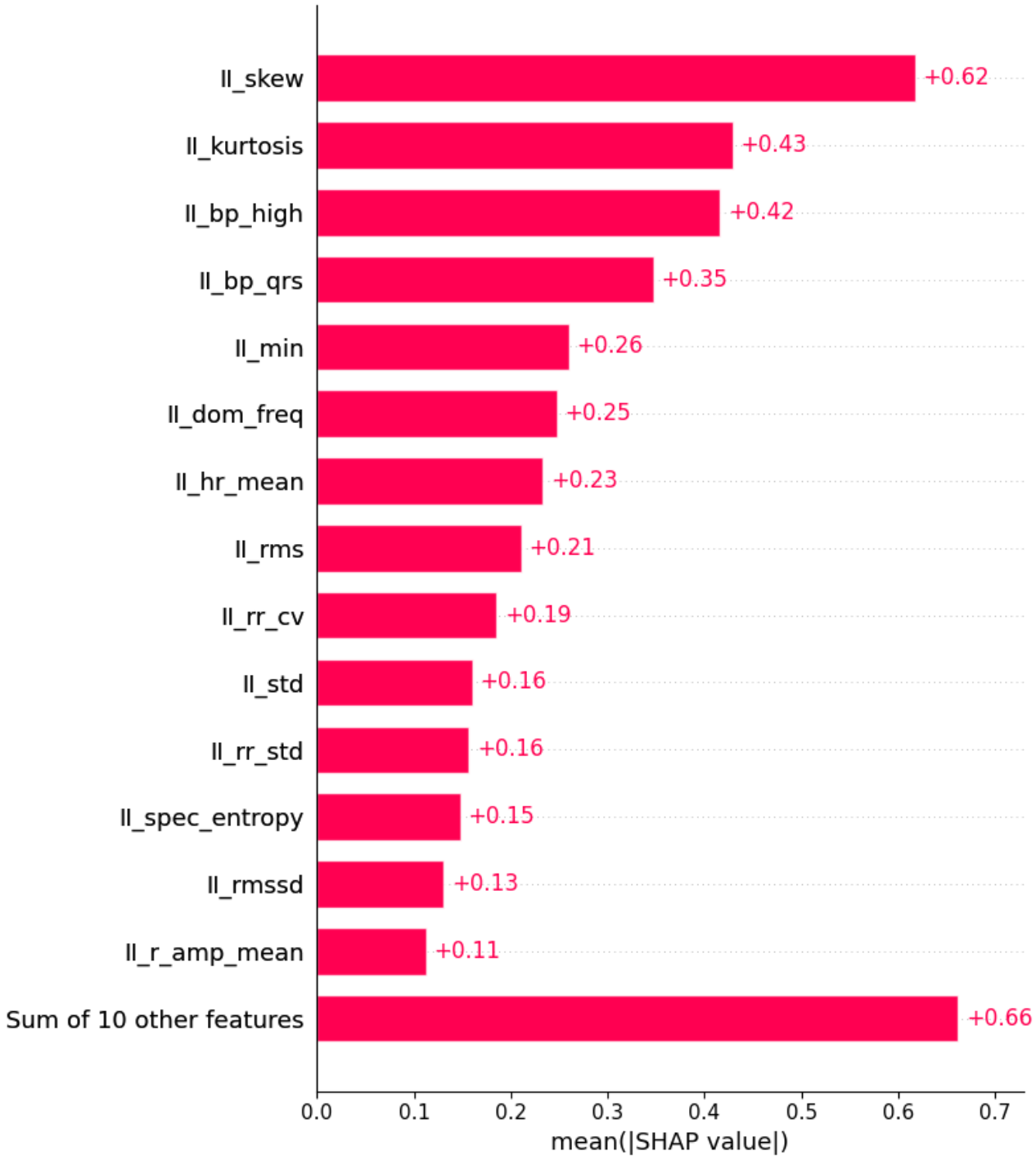
	n_test_records	recall
MI	385	0.725
STTC	357	0.711
CD	345	0.748
HYP	189	0.688

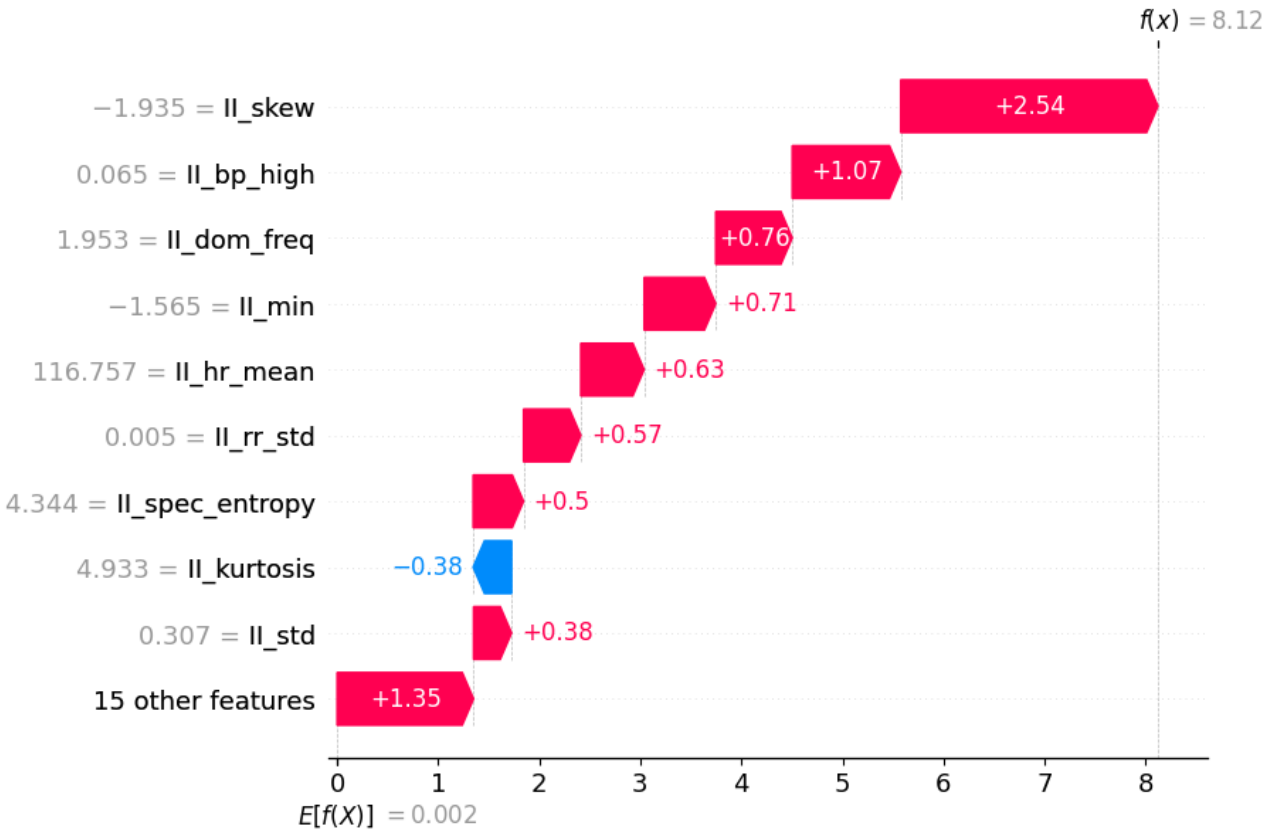
False positives: 117 | False negatives: 289
Mean feature values of truly abnormal records, by predicted class (0 = missed, 1 = detected):

	II_hr_mean	II_rr_cv	II_range	II_rms	II_bp_qrs
pred					
0	73.106	0.065	1.098	0.134	0.439
1	78.126	0.107	1.078	0.131	0.422

Explainability




II_skew
II_kurtosis
II_bp_high
II_bp_qrs
II_min
II_dom_freq
II_hr_mean
II_rms
II_rr_cv
II_std
II_rr_std
II_spec_entropy
II_rmssd
II_r_amp_mean
Sum of 10 other features
+0.62
+0.43
+0.42
+0.35
+0.26
+0.25
+0.23
+0.21
+0.19
+0.16
+0.16
+0.15
+0.13
+0.11
+0.66
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
mean(|SHAP value|)



Permutation Importance

